MnVisionAI: Space-Intelligence Platform for Manganese Reserve Discovery

Comprehensive SIH 2026 PS SIH26009 Implementation Plan, Architecture and Roadmap

Problem Understanding

Use AI/ML plus satellite, geological and mining data to identify prospective manganese zones, rank exploration targets and reduce production shortfalls. Detailed brainstorming: combine spectral signatures, terrain derivatives, lineament density, hydrothermal alteration indicators, clustering of historical deposits, and graph-based spatial relationships to improve discovery accuracy. Establish feedback loop from field surveys to continuously retrain the model.

Project Vision

Create a national decision platform combining remote sensing, GIS, predictive AI, explainable analytics and exploration planning. Detailed brainstorming: combine spectral signatures, terrain derivatives, lineament density, hydrothermal alteration indicators, clustering of historical deposits, and graph-based spatial relationships to improve discovery accuracy. Establish feedback loop from field surveys to continuously retrain the model.

Key Innovations

Multi-source data fusion; geospatial foundation model; explainable mineral prospectivity maps; uncertainty scoring; digital twin of exploration. Detailed brainstorming: combine spectral signatures, terrain derivatives, lineament density, hydrothermal alteration indicators, clustering of historical deposits, and graph-based spatial relationships to improve discovery accuracy. Establish feedback loop from field surveys to continuously retrain the model.

Architecture

Data Layer: Sentinel/Landsat/ASTER + geology maps + DEM + mine records. Processing Layer: ETL, raster processing, feature engineering. AI Layer: Random Forest, XGBoost, CNNs. Services Layer: APIs, GIS engine. Presentation Layer: Web dashboard and exploration planner. Detailed brainstorming: combine spectral signatures, terrain derivatives, lineament density, hydrothermal alteration indicators, clustering of historical deposits, and graph-based spatial relationships to improve discovery accuracy. Establish feedback loop from field surveys to continuously retrain the model.

Datasets

Sentinel-2, Landsat-8/9, ASTER, SRTM DEM, Geological Survey maps, mineral inventory records, soil and geomorphology layers. Detailed brainstorming: combine spectral signatures, terrain derivatives, lineament density, hydrothermal alteration indicators, clustering of historical deposits, and graph-based spatial relationships to improve discovery accuracy. Establish feedback loop from field surveys to continuously retrain the model.

AI Pipeline

Image preprocessing, spectral indices, lithology extraction, anomaly detection, supervised prospectivity modeling, ensemble ranking, validation. Detailed brainstorming: combine spectral signatures, terrain derivatives, lineament density, hydrothermal alteration indicators, clustering of historical deposits, and graph-based spatial relationships to improve discovery accuracy. Establish feedback loop from field surveys to continuously retrain the model.

Tech Stack

Python, GeoPandas, Rasterio, QGIS, PostGIS, FastAPI, PyTorch, XGBoost, React, Leaflet/Mapbox. Detailed brainstorming: combine spectral signatures, terrain derivatives, lineament density, hydrothermal alteration indicators, clustering of historical deposits, and graph-based spatial relationships to improve discovery accuracy. Establish feedback loop from field surveys to continuously retrain the model.

Implementation Plan

Phase 1 Data Collection. Phase 2 Data Lake. Phase 3 ML Prototypes. Phase 4 GIS Platform. Phase 5 Field Validation. Phase 6 Optimization. Detailed brainstorming: combine spectral signatures, terrain derivatives, lineament density, hydrothermal alteration indicators, clustering of historical deposits, and graph-based spatial relationships to improve discovery accuracy. Establish feedback loop from field surveys to continuously retrain the model.

Hackathon MVP

Upload region, generate manganese probability heatmap, top candidate zones, confidence scores and downloadable report. Detailed brainstorming: combine spectral signatures, terrain derivatives, lineament density, hydrothermal alteration indicators, clustering of historical deposits, and graph-based spatial relationships to improve discovery accuracy. Establish feedback loop from field surveys to continuously retrain the model.

6 Month Roadmap

Month1 data ingestion. Month2 feature engineering. Month3 model training. Month4 dashboard. Month5 validation. Month6 deployment and scaling. Detailed brainstorming: combine spectral signatures, terrain derivatives, lineament density, hydrothermal alteration indicators, clustering of historical deposits, and graph-based spatial relationships to improve discovery accuracy. Establish feedback loop from field surveys to continuously retrain the model.

Team Structure

AI Lead, GIS Lead, Backend Developer, Frontend Developer, Domain Researcher, Presentation Lead. Detailed brainstorming: combine spectral signatures, terrain derivatives, lineament density, hydrothermal alteration indicators, clustering of historical deposits, and graph-based spatial relationships to improve discovery accuracy. Establish feedback loop from field surveys to continuously retrain the model.

Judging Advantages

High societal impact, strong innovation, measurable outcomes, scalable architecture and explainable AI. Detailed brainstorming: combine spectral signatures, terrain derivatives, lineament density, hydrothermal alteration indicators, clustering of historical deposits, and graph-based spatial relationships to improve discovery accuracy. Establish feedback loop from field surveys to continuously retrain the model.

Detailed System Flow

Satellite Data -> Preprocessing -> Feature Extraction -> Mineral Prospectivity Engine -> Ensemble AI Models -> Explainability Layer -> GIS Dashboard -> Exploration Recommendations.

Expected Outputs: probability maps, reserve hotspots, drill-priority rankings, risk indicators, production forecasting and policy insights.